

MANIPAL
ACADEMY of HIGHER EDUCATION
(Institution of Eminence Deemed to be University)

MANIPAL SCHOOL OF INFORMATION SCIENCES

(A Constituent unit of MAHE, Manipal)

CYE 5153 Linux OS and Scripting Lab – 6.1

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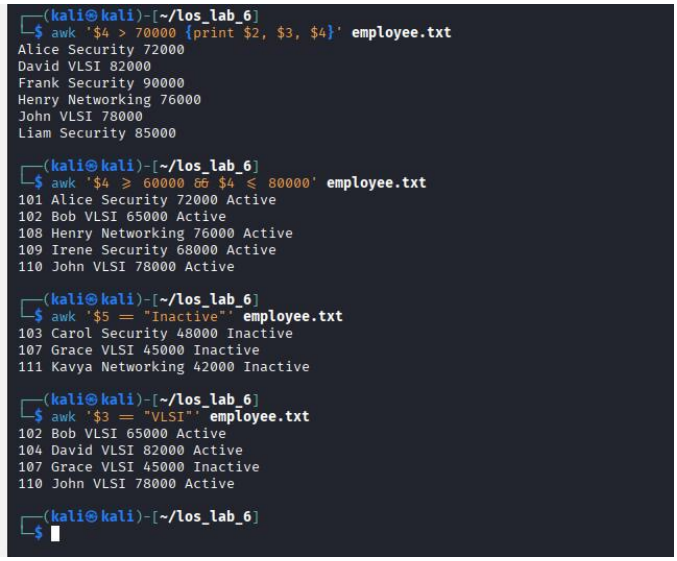
Part A Numerical and String Comparisons

Display the name, department, and salary of employees earning more than 70000.

Display employees whose salary is between 60000 and 80000, including both limits.

Display employees whose employment status is Inactive.

Display employees belonging to the VLSI department.



Part B Regular-Expression Patterns

Display all records containing the word Security.

Display employees whose names begin with the letters A, B, or C.

Apply the regular expression only to the employee-name field.

Display employees whose department field does not match Security.

```
┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '/Security/' employee.txt
101 Alice Security 72000 Active
103 Carol Security 48000 Inactive
106 Frank Security 90000 Active
109 Irene Security 68000 Active
112 Liam Security 85000 Active

┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '$2 ~ /^[ABC]/' employee.txt
101 Alice Security 72000 Active
102 Bob VLSI 65000 Active
103 Carol Security 48000 Inactive

┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '$3 !~ /Security/' employee.txt
102 Bob VLSI 65000 Active
104 David VLSI 82000 Active
105 Eva Networking 55000 Active
107 Grace VLSI 45000 Inactive
108 Henry Networking 76000 Active
110 John VLSI 78000 Active
111 Kavya Networking 42000 Inactive

┌──(kali㉿kali)-[~/los_lab_6]
└─$
```

Part C Logical Patterns

Display employees who work in the Security department and earn more than 70000.

Display employees who belong to either the VLSI or Networking department.

Display employees who: Work in the VLSI department, Earn more than 70000, have Active status

```
┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '$3 == "Security" && $4 > 70000' employee.txt
101 Alice Security 72000 Active
106 Frank Security 90000 Active
112 Liam Security 85000 Active

┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '$3 == "VLSI" || $3 == "Networking"' employee.txt
102 Bob VLSI 65000 Active
104 David VLSI 82000 Active
105 Eva Networking 55000 Active
107 Grace VLSI 45000 Inactive
108 Henry Networking 76000 Active
110 John VLSI 78000 Active
111 Kavya Networking 42000 Inactive

┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '$3 == "VLSI" && $4 > 70000 && $5 == "Active"' employee.txt
104 David VLSI 82000 Active
110 John VLSI 78000 Active

┌──(kali㉿kali)-[~/los_lab_6]
└─$
```

Part D Conditional Pattern

Department-dependent salary condition

Apply different salary conditions:

If the employee belongs to Security, select employees earning more than 75000.

Otherwise, select employees earning more than 70000.

Display the employee name, department, and salary.

```
┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '($3 == "Security" && $4 > 75000) || ($3 != "Security" && $4 > 70000) {print $2, $3, $4}' employee.txt
David VLSI 82000
Frank Security 90000
Henry Networking 76000
John VLSI 78000
Liam Security 85000
```

Part E Grouped Patterns

Display active employees who belong to either the Security or VLSI department.

The Active condition must apply to both departments.

Display employees who satisfy one of these conditions:

Work in Security and earn more than 80000

Work in VLSI and earn more than 75000

Display active employees who satisfy one of these conditions:

Work in Security and earn more than 70000

Work in VLSI and earn more than 75000

```
┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '$5 == "Active" && ($3 == "Security" || $3 == "VLSI")' employee.txt
101 Alice Security 72000 Active
102 Bob VLSI 65000 Active
104 David VLSI 82000 Active
106 Frank Security 90000 Active
109 Irene Security 68000 Active
110 John VLSI 78000 Active
112 Liam Security 85000 Active

┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '($3 == "Security" && $4 > 80000) || ($3 == "VLSI" && $4 > 75000)' employee.txt
104 David VLSI 82000 Active
106 Frank Security 90000 Active
110 John VLSI 78000 Active
112 Liam Security 85000 Active

┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '$5 == "Active" && (($3 == "Security" && $4 > 70000) || ($3 == "VLSI" && $4 > 75000))' employee.txt
101 Alice Security 72000 Active
104 David VLSI 82000 Active
106 Frank Security 90000 Active
110 John VLSI 78000 Active
112 Liam Security 85000 Active

┌──(kali㉿kali)-[~/los_lab_6]
└─$
```

Part F Negated Patterns

Display employees whose status is not Active.

Use an exact field comparison because the word Inactive contains the text Active.

Display employees who do not belong to either the Security or VLSI department.

```
┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '$5 ≠ "Active"' employee.txt
103 Carol Security 48000 Inactive
107 Grace VLSI 45000 Inactive
111 Kavya Networking 42000 Inactive

┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '$3 ≠ "Security" && $3 ≠ "VLSI"' employee.txt
105 Eva Networking 55000 Active
108 Henry Networking 76000 Active
111 Kavya Networking 42000 Inactive

┌──(kali㉿kali)-[~/los_lab_6]
└─$
```

Part G Range Patterns

Display all records beginning with employee ID 103 and ending with employee ID 108. Include both boundary records.

Display all records beginning with David and ending with Irene. Include both boundary records.

```
┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '$1 == 103, $1 == 108' employee.txt
103 Carol Security 48000 Inactive
104 David VLSI 82000 Active
105 Eva Networking 55000 Active
106 Frank Security 90000 Active
107 Grace VLSI 45000 Inactive
108 Henry Networking 76000 Active

┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '$2 == "David", $2 == "Irene"' employee.txt
104 David VLSI 82000 Active
105 Eva Networking 55000 Active
106 Frank Security 90000 Active
107 Grace VLSI 45000 Inactive
108 Henry Networking 76000 Active
109 Irene Security 68000 Active

┌──(kali㉿kali)-[~/los_lab_6]
└─$
```

Part H Built-In Variables

Display only the first five records. Include the record number in the output.

Display records numbered 4 through 9 using the built-in variable NR.

Display only records containing exactly five fields. Print the record number and the complete record.

Display only odd-numbered records using NR.



Part I Arithmetic Conditions

Assume each employee receives a bonus equal to 10% of the salary. Display employees whose calculated bonus exceeds 7000.

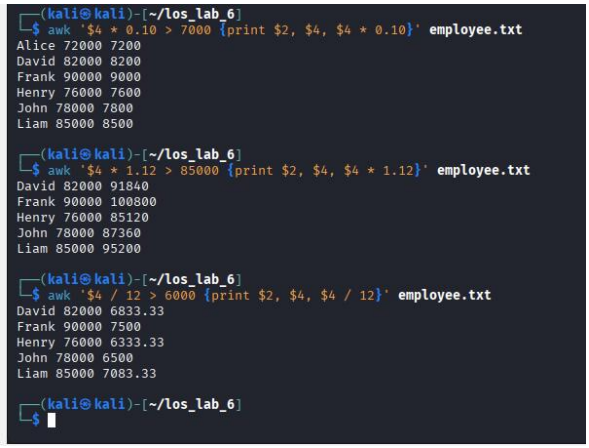
Print: Employee name, Salary, Calculated bonus

Question 24 Revised salary

Calculate the revised salary after adding a 12% increment. Display employees whose revised salary exceeds 85000.

=

Calculate the monthly salary by dividing the annual salary by 12. Display employees whose monthly salary exceeds 6000.



Part J Integrated Challenge

Generate a report containing active employees who satisfy one of these conditions:

Work in Security and earn at least 70000, Work in VLSI and earn at least 75000, Work in Networking and earn at least 70000

The output must contain: Employee ID, Employee name, Department, Salary, Status

Add a report heading using BEGIN and print the total number of selected employees using END.

```
┌──(kali㉿kali)-[~/los_lab_6]
└─$ awk '
BEGIN {
    print "═══ SELECTED ACTIVE EMPLOYEES REPORT ═══"
    printf "%-5s %-10s %-12s %-8s %-8s\n", "ID", "Name", "Department", "Salary", "Status"
    print "──────────────────────────────────────────────"
}
$5 == "Active" && (($3 == "Security"   && $4 >= 70000) ||
                   ($3 == "VLSI"       && $4 >= 75000) ||
                   ($3 == "Networking" && $4 >= 70000)) {
    printf "%-5s %-10s %-12s %-8s %-8s\n", $1, $2, $3, $4, $5
    count++
}
END {
    print "──────────────────────────────────────────────"
    print "Total selected employees:", count
}' employee.txt
═══ SELECTED ACTIVE EMPLOYEES REPORT ═══
ID    Name       Department   Salary   Status
──────────────────────────────────────────────
101   Alice      Security     72000    Active
104   David      VLSI         82000    Active
106   Frank      Security     90000    Active
108   Henry      Networking   76000    Active
110   John       VLSI         78000    Active
112   Liam       Security     85000    Active
──────────────────────────────────────────────
Total selected employees: 6
```